3B Profit, Loss, Discounts and Simple Interest

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Calculate the profit or loss and the percentage profit or loss on cost price:

(a) Cost \$60, Sold for \$75 (b) Cost \$120, Sold for \$96 (c) Cost \$80, Sold for \$116 (d) Cost \$250, Sold for \$215

Q2. Find the selling price after each mark-up or discount:

(a) \$90 marked up by 30% (b) \$160 discounted by 15% (c) \$240 marked up by 35% (d) \$350 discounted by 28%

Q3. Calculate the simple interest $I = Prt$ earned on each investment:

(a) $P = \$2000, r = 5\%$ p.a., $t = 3$ years

(b) $P = \$4500, r = 4\%$ p.a., $t = 5$ years

(c) $P = \$8000, r = 4.5\%$ p.a., $t = 2$ years

(d) $P = \$3600, r = 6\%$ p.a., $t = 6$ months (0.5 yr)

Q4. Use the simple interest formula $I = Prt$ and $A = P + I$ to find the required value:

(a) Total amount A when $P = \$5000, r = 4.8\%$ p.a., $t = 3$ yr

(b) Time t (years) when $P = \$3000, r = 5\%$ p.a., $I = \$600$

(c) Rate r (% p.a.) when $P = \$4000, t = 3$ yr, $I = \$540$

(d) Principal P when $r = 6\%$ p.a., $t = 4$ yr, $I = \$600$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A surf shop in Torquay buys wetsuits at a wholesale cost of \$150 each and marks them up by 60% to set the retail price. During an end-of-summer sale, the shop offers 25% off the retail price. Find (a) the retail price, (b) the sale price, and (c) the percentage profit still made on the \$150 wholesale cost.

Q6. Chloe invests \$6500 in a Bendigo Bank term deposit paying simple interest of 5.2% p.a. for 3 years. Find (a) the total simple interest earned and (b) the total value of her investment at maturity.

Q7. A real estate agent in Geelong charges a commission of 2.2% on the selling price of a house. If a house sells for \$750 000, how much commission does the agent earn, and how much does the vendor receive after commission?

Q8. An electronics retailer advertises a \$1200 laptop with a 20% Boxing Day discount, plus an extra 10% off the discounted price for students. Find the final price paid by a student and the single equivalent percentage discount.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. At a simple interest rate of $r = 8\%$ p.a., how many years t does it take for any principal P to double in value (so that the simple interest earned equals the original principal, $I = P$)? Show why the answer does not depend on P.

Q10. Store A applies a 30% discount followed by a 10% discount. Store B applies a 10% discount followed by a 30% discount. Prove using decimal multipliers that both stores charge the exact same final price.